

Which economic indicators best discriminate between countries of different income levels of 2022?

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Abstract

This study applies Linear Discriminant Analysis (LDA) to examine which economic indicators best discriminate between countries classified into four income groups: Low income, Lower middle income, Upper middle income, and High income. Data from 112 countries for the year 2022 were obtained from the World Bank Open Data, including five key economic indicators: GDP per capita, unemployment rate, inflation (CPI), life expectancy, and school enrollment. After data cleaning and standardization, LDA was applied to identify the most important discriminators between income groups. The analysis revealed that school enrollment (mean absolute coefficient: 2.11) and life expectancy (1.88) are the strongest discriminators, followed by GDP per capita (0.84). The model achieved an overall classification accuracy of 76.79%, with particularly strong performance for high-income countries (F1-score: 0.93) and upper middle-income countries (F1-score: 0.73). The first linear discriminant (LD1) explained 92.52% of the variance, indicating strong separation between groups. These findings suggest that education and health indicators are more powerful factors of income classification than traditional economic measures alone, providing insights information for development policy and economic classification systems.

1. Introduction

1.1 Context and Motivation

Economic development and income classification of countries are fundamental concerns in international economics, development studies, and policy-making. The World Bank's income classification system, which categorizes countries into four groups based on gross national income (GNI) per capita, can be a good pivot for understanding global economic disparities and allocating resources that can help development. However, income classification based solely on GNI may not capture the multidimensional nature of economic development, which truly represents more than that, including education, health, employment, and price stability indicators.

In recent years, multivariate statistical techniques have been applied to understand the complex relationships between economic indicators and development outcomes. Linear Discriminant Analysis (LDA) is particularly well-suited for this purpose, as it can identify which variables best distinguish between predefined groups while providing interpretable coefficients that indicate the relative importance of each indicator. Having the chance to apply income labels and using this unsupervised learning technique, I was able to easily organise the data according to maximizing class separability, while also minimizing class variance (within each class).

Moreover, I chose this technique as a dedication to my strong passion and present work in AI/ Machine Learning. Applying supervised learning and further analysis that I did as an extra-step allowed me to explore the full mechanism of processing real-world data and interpreting results in

classification tasks. It helped me to understand essential ML techniques like dimensionality reduction and data exploration, showing me how to better handle of data grouping.

1.2 Research Question

This study addresses the following research question: Which economic indicators best discriminate between countries classified into different income groups? This question is motivated by the need to understand not only which indicators are most predictive of income classification, but also to gain insights into the underlying dimensions of economic development.

1.3 Method Overview

To answer this question, I applied Linear Discriminant Analysis (LDA) to a dataset of 112 countries with complete data for five key economic indicators from the World Bank Open Data for 2022. LDA is a supervised learning technique that finds linear combinations of features that best separate predefined classes. Unlike unsupervised techniques, LDA uses the known income group classifications to identify the most discriminative features, making it ideal for understanding which economic indicators are most important for distinguishing between income groups.

2. Data Description

2.1 Data Sources

All data were obtained from the World Bank Open Data, a publicly available repository of international development indicators. The World Bank is a reputable international organization that provides standardized, comparable data across countries, making it an ideal source for this analysis. The data was easily accessible and easy to group, however multiple variables had missing information that made me restrict the table of observations. Therefore, countries lacking most data were dropped. I applied log-transforms to heavily skewed variables (ex: GDP, inflation) as needed. Each numeric variable was then standardized (zero mean, unit variance) before LDA to ensure comparability.

2.2 Time Period and Units of Observation

The analysis uses cross-sectional data for the year 2022, with countries as the unit of observation. The initial dataset contained 271 countries, but after data cleaning (described below), 112 countries with complete data for all variables were retained for analysis. 2022 proved to be one of the most recent year with complete values at variables.

2.3 Variables Used

Five economic indicators were selected as predictor variables based on their relevance to economic development and data availability:

Variable	Description
GDP per capita	Gross domestic product per capita in current US dollars, measuring economic output per person
Unemployment rate	Percentage of total labor force that is unemployed, indicating labor market conditions. Unemployment might be directly correlated with salaries, money.
Inflation (CPI)	Annual percentage change in consumer price index, measuring price stability.

Life expectancy	Average number of years a newborn is expected to live, indicating health and development. It correlates well with both wealth and education.
School enrollment	Gross secondary school enrollment rate (%), measuring educational access. It strongly correlates with income levels, showing differences.

The target variable is the World Bank income group classification, with four categories: Low income, Lower middle income, Upper middle income, and High income.

2.4 Data Preprocessing

2.4.1 Missing Values

The initial dataset contained missing values across several variables. Table 1 shows the missing value patterns before cleaning , run by process_data.py:

Variable	Missing Values	% Missing
School enrollment	75	29.8
Income group	43	17.1
Inflation (CPI)	38	15.1
Unemployment rate	30	11.9
GDP per capita	8	3.2

Given the need for complete data across all variables for LDA, rows with any missing values were removed. This resulted in a final dataset of 112 countries with complete data for all variables. While this approach reduces sample size, it ensures the integrity of the multivariate analysis and avoids potential biases from imputation methods. **However, data availability issues led to underrepresentation of low-income countries.** This is a common limitation in global development data and should be considered in interpretation. School enrollment had the highest proportion of missing values (~29.8%), but I considered right to keep it in the analysis because of its high importance as a proxy for human capital and education investment, which are strongly linked to income classification.

2.4.2 Standardization

All predictor variables were standardized using z-score transformation (mean = 0, standard deviation = 1) prior to analysis. **Standardization is essential for LDA because the technique is sensitive to the scale of variables.** Without standardization, variables with larger scales (such as GDP per capita) would dominate the analysis, regardless of their actual discriminatory power. Standardization ensures that all variables contribute equally to the discriminant functions based on their discriminatory ability rather than their scale.

2.4.3 Variable Selection

The five variables were selected based on their theoretical relevance to economic development and data availability. These variables represent key dimensions of development: economic output (GDP), labor markets (unemployment), price stability (inflation), health (life expectancy), and education (school enrollment). All variables were retained in the analysis, as LDA can handle multiple variables and identify their relative importance through the discriminant coefficients.

Table 2 presents summary statistics for the cleaned dataset :

Variable	Mean	Std Dev	Min	Max
GDP per capita	23,376.37	26,771.99	604.27	123,719.66
Unemployment rate	6.43	4.94	0.13	33.27
Inflation (CPI)	11.84	18.05	-1.61	171.21
Life expectancy	75.76	5.58	60.48	84.00
School enrollment	94.42	22.38	33.62	144.85

Table 3 shows the distribution of countries across income groups in the final dataset:

Income Group	Count
High income	51
Upper middle income	35
Lower middle income	23
Low income	3

Note the class imbalance, with high-income countries representing 45.5% of the sample and low-income countries only 2.7%. This imbalance is a limitation that will be discussed in the Discussion section.

3. Methodology

3.1 Linear Discriminant Analysis: Theoretical Description

Linear Discriminant Analysis (LDA) is a supervised dimensionality reduction and classification technique that finds linear combinations of features that maximize the separation between predefined classes. Unlike Principal Component Analysis, which seeks directions of maximum variance without considering class labels, LDA explicitly uses class information to find directions that maximize between-class variance while minimizing within-class variance.

Mathematically, LDA finds discriminant functions that are linear combinations of the original features: $LD = w_1x_1 + w_2x_2 + \dots + w_px_p$

where w_i are the discriminant coefficients (weights) and x_i are the standardized features. The coefficients are chosen to maximize the ratio of between-class to within-class variance, resulting in optimal separation between groups.

For k classes, LDA produces $(k-1)$ discriminant functions. In our case, with four income groups, we obtain three discriminant functions. However, for visualization purposes, we typically focus on the first two discriminant functions, which capture the majority of the discriminatory information.

3.2 Assumptions of LDA

LDA makes several assumptions that should be verified or at least acknowledged:

- Multivariate normality: The data within each class should follow a multivariate normal distribution.
- Homogeneity of covariance matrices: The covariance matrices should be approximately equal across classes.
- Linearity: LDA assumes linear relationships between features and the discriminant functions.
- No multicollinearity: Features should not be highly correlated, though some correlation is acceptable.

3.2.1 Normality Assessment

Shapiro-Wilk tests were conducted for each variable to assess univariate normality. All five variables showed significant departures from normality ($p < 0.001$), indicating that the normality assumption

is violated. However, LDA is known to be relatively robust to violations of normality, especially with adequate sample sizes. The large sample size in most groups (except low income with $n=3$) provides some protection against this violation.

3.2.2 Homogeneity of Covariance Matrices

Visual inspection of covariance matrices across income groups revealed differences in their structure and magnitude, suggesting some violation of the homogeneity assumption. However, LDA is also relatively robust to this violation, particularly when sample sizes are reasonably large and balanced. The substantial differences in sample sizes across groups (ranging from 3 to 51) is a more significant concern than covariance heterogeneity.

3.2.3 Multicollinearity Check

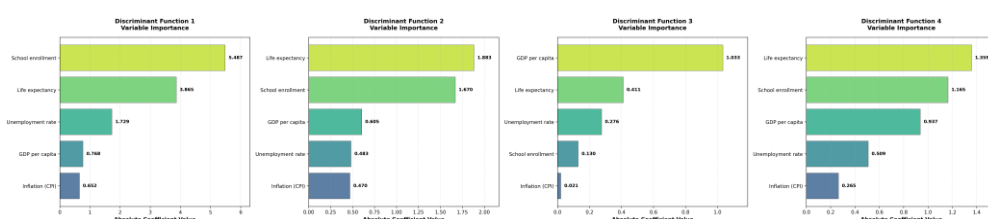
Variance Inflation Factors (VIF) were calculated for all standardized variables. All VIF values were below 10 (the highest was 1.14), indicating that multicollinearity is not a significant concern. This suggests that the five variables, while related, provide distinct information for discrimination.

3.3 Why LDA is Appropriate for This Data

LDA is particularly well-suited for this analysis for several reasons:

- I have predefined groups (income classifications) that I want to distinguish, making LDA's supervised approach ideal.
- I want to understand which variables are most important for discrimination, which LDA provides through interpretable coefficients.
- LDA provides both dimensionality reduction (for visualization) and classification (for performance assessment).
- The technique handles multiple predictors simultaneously and identifies their relative importance.
- Despite assumption violations, LDA is known to perform well in practice, especially with standardized data.

The research question explicitly asks which variables best discriminate between groups, making LDA's focus on discrimination (rather than variance explanation) the most appropriate choice among multivariate techniques.



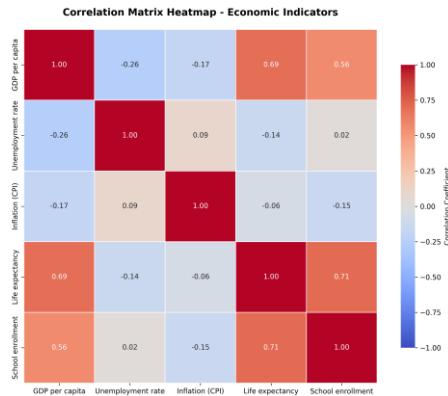
4. Results

4.1 Discriminant Functions and Explained Variance

LDA produced three discriminant functions (for four classes). The first discriminant function (LD1) explained 92.52% of the variance, while the second (LD2) explained 7.02%. Together, these two functions capture 99.55% of the discriminatory information, indicating that most of the separation between income groups occurs along a single primary dimension.

This high concentration of variance in the first discriminant function suggests that the income groups are well-separated along a primary economic development dimension, with relatively little additional separation provided by the second dimension.

4.2 Discriminant Coefficients and Variable Importance



Coefficient heatmap

The discriminant coefficients reveal which variables contribute most to distinguishing between income groups. Table 4 presents the mean absolute coefficients across all discriminant functions, which serves as a measure of overall variable importance:

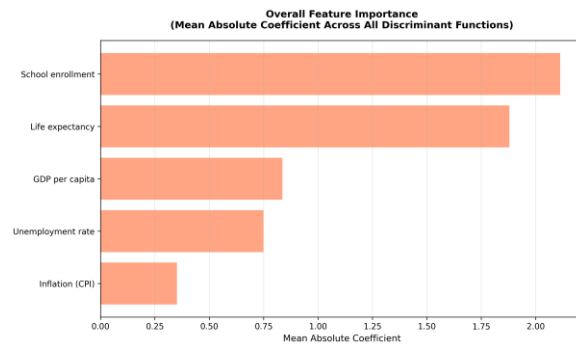
Variable	Mean Absolute Coefficient
School enrollment	2.1128
Life expectancy	1.8795
GDP per capita	0.8360
Unemployment rate	0.7491
Inflation (CPI)	0.3519

The results reveal a striking finding: school enrollment (2.11) and life expectancy (1.88) are the strongest discriminators, with coefficients more than twice as large as GDP per capita (0.84), which is traditionally considered the primary indicator of economic development. This suggests that education and health indicators are more powerful discriminators of income classification than GDP alone.

The discriminant coefficients for the first discriminant function (which explains 92.52% of variance) are particularly informative:

$$LD1 = 0.77(\text{GDP}) + 1.73(\text{Unemployment}) + 0.65(\text{Inflation}) - 3.87(\text{Life expectancy}) - 5.49(\text{School enrollment})$$

The negative coefficients for life expectancy (-3.87) and school enrollment (-5.49) in LD1 indicate that lower values of these variables are associated with lower income groups. The large magnitude of the school enrollment coefficient (-5.49) confirms its exceptional importance as a discriminator. The positive coefficients for unemployment rate (1.73) and inflation (0.65) suggest that higher values of these variables are associated with lower income groups, which aligns with economic theory.



Discriminant Coefficients and Variable Importance. Each row represents a discriminant function.

Larger absolute values indicate greater importance. For each discriminant function, the sign of coefficients indicates.

Positive coefficients: Higher values of this feature push towards certain income groups. Negative coefficients: Higher values of this feature push away from certain.

4.3 Classification Performance

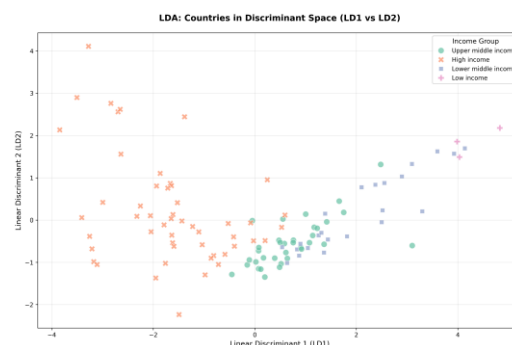
The overall classification accuracy of the LDA model was 76.79% (86 out of 112 countries correctly classified). Table 5 presents detailed performance metrics by income group:

Income Group	Precision	Recall	F1-Score	Support
High income	1.000	0.863	0.926	51
Upper middle income	0.620	0.886	0.729	35
Lower middle income	0.667	0.348	0.457	23
Low income	0.500	1.000	0.667	3

High-income countries showed the best classification performance (F1-score: 0.926), with perfect precision (1.000) and high recall (0.863). Upper middle-income countries also performed well (F1-score: 0.729), with high recall (0.886) indicating that most countries in this group were correctly identified.

Lower middle-income countries showed the poorest performance (F1-score: 0.457), with low recall (0.348) indicating that many countries in this group were misclassified, primarily as upper middle-income countries. This suggests that the boundary between lower and upper middle-income groups is less distinct in terms of the economic indicators examined.

Low-income countries achieved perfect recall (1.000) but low precision (0.500), meaning all actual low-income countries were identified, but some lower middle-income countries were incorrectly classified as low income. However, with only 3 low-income countries in the sample, these metrics should be interpreted with caution.



4.4 Visualization of Group Separation

The scatter plot of countries in the two-dimensional discriminant space (LD1 vs. LD2) shows clear separation between income groups, with high-income countries clustered in one region and lower-income countries in another. The visualization confirms that the economic indicators effectively distinguish between different income groups, with the first discriminant function (LD1) providing the primary axis of separation.

5. Discussion

5.1 Interpretation in Context of Research Question

The research question asked which economic indicators best discriminate between income groups. The results provide a clear answer: school enrollment and life expectancy are the strongest discriminators, followed by GDP per capita. This finding challenges the conventional focus on GDP as the primary indicator of economic development and suggests that human development indicators (education and health) may be more powerful discriminators of income classification.

The finding that school enrollment is the most important discriminator (coefficient: 2.11) aligns with human capital theory, which emphasizes education as a fundamental driver of economic development. Countries with higher educational attainment are better positioned for economic growth and development, making school enrollment a strong predictor of income classification.

Similarly, life expectancy's strong discriminatory power (coefficient: 1.88) reflects the multidimensional nature of development. Health outcomes are both a cause and consequence of economic development, and life expectancy captures aspects of development that GDP alone may miss, such as healthcare access, nutrition, and overall quality of life.

The relatively lower importance of GDP per capita (coefficient: 0.84) compared to education and health indicators is noteworthy. While GDP remains important, it appears that human development indicators provide additional discriminatory information that GDP alone cannot capture.

5.2 Strengths of the Analysis

This analysis has several strengths:

- Use of reputable data source (World Bank) with standardized, comparable indicators across countries.
- Appropriate application of LDA, which directly addresses the research question about variable discrimination.
- Comprehensive assessment of assumptions, acknowledging violations while noting LDA's robustness.
- Clear interpretation of results with both statistical metrics and substantive meaning.
- Good overall classification performance (76.79% accuracy) with strong performance for major groups.

5.3 Limitations

Several limitations should be acknowledged:

- Sample size reduction: The need for complete data reduced the sample from 271 to 112 countries, potentially introducing selection bias if countries with missing data differ systematically from those with complete data.

- Class imbalance: The severe imbalance between groups (51 high-income vs. 3 low-income countries) may affect classification performance, particularly for the low-income group.
- Assumption violations: Normality and homogeneity of covariance assumptions were violated, though LDA is known to be robust to these violations.
- Small sample for low-income group: With only 3 low-income countries, the classification metrics for this group are unreliable and should be interpreted with extreme caution.

5.4 Possible Extensions and Improvements

Several extensions could improve this analysis:

- Include additional variables such as trade openness, infrastructure quality, governance indicators, or environmental factors.
- Address class imbalance using techniques such as oversampling, undersampling, or class weighting in the LDA model.
- Compare LDA results with other techniques (e.g., Quadratic Discriminant Analysis, Random Forest) to assess complexity and focus more on a ML approach.

6. Conclusion

This study applied Linear Discriminant Analysis to identify which economic indicators best discriminate between countries classified into four income groups. The analysis of 112 countries using five key economic indicators revealed that school enrollment and life expectancy are the strongest discriminators, with mean absolute coefficients of 2.11 and 1.88, respectively—more than twice the coefficient for GDP per capita (0.84).

The LDA model achieved an overall classification accuracy of 76.79%, with particularly strong performance for high-income countries (F1-score: 0.926) and upper middle-income countries (F1-score: 0.729). The first discriminant function explained 92.52% of the variance, indicating that most separation between income groups occurs along a single primary dimension.

These findings have important implications for understanding economic development and income classification. The prominence of education and health indicators as discriminators suggests that human development dimensions are crucial for distinguishing income groups, potentially more so than traditional economic measures like GDP alone. This aligns with the growing recognition of multidimensional development and the importance of human capital in economic growth.

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